

Peter Yelton

Fullstack Developer | Design Engineer |

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Design engineer with fullstack and AI implementation experience. I focus on building user-facing features that are both technically solid and visually compelling - I handle everything from the backend logic to the polished frontend experience

Core Technical Skills

- **Development:** TypeScript, React, Next.js, Express.js, Python Flask, React Native/Expo
- **AI Integration:** AI SDK, Conversational AI, Real-time Chat Systems
- **Design Tools:** Figma, Adobe Creative Suite, Blender, UI/UX Design
- **Leadership:** Department Strategy, Cross-functional Collaboration, Production Optimization

Featured Projects

- **Mise - AI-Powered Recipe Platform** [link](#)
 - Built fullstack application using TypeScript, React, and Next.js with Expo/React Native mobile port
 - Integrated AI SDK for conversational recipe generation with real-time dietary customization and substitutions
 - Developed Express.js backend API supporting both web and mobile platforms
- **Mise - Expo App**
 - Developed complete 2D game featuring office task management gameplay mechanics
 - Created all visual art assets and character animations using digital design tools
 - Implemented game logic and user interface demonstrating end-to-end creative and technical execution
- **Peter's World** [link](#)
 - **Living Ecosystem Simulation:** A small 3D world featuring autonomous animals that graze on grass and trees that naturally grow, spread, and die over time, creating a dynamic environment that evolves without player intervention.
 - **Interactive Terraforming:** Players can reshape the landscape using various terraforming tools, allowing them to modify terrain, plant vegetation, and influence how the ecosystem develops and thrives.
 - **Physics-Based World:** Built as a React Next.js application using React Three Fiber for 3D rendering and Rapier.js for realistic gravity and physics simulation, ensuring natural movement and interactions between all world elements.

EXPERIENCE

Fractal Tech AI Accelerator | Remote

Software Engineering Fellow | June 2025 - August 2025

Intensive 60-hour/week accelerator program focused on rapid AI application development:

- **High-velocity shipping:** Delivered 3-5 pull requests daily across multiple full-stack TypeScript projects, building production-ready features with Next.js and React while maintaining code quality
- **AI-assisted development workflow:** Leveraged AI SDK and modern development tools to accelerate feature implementation, establishing systematic approaches to fast iteration without accumulating technical debt
- **Production deployment cadence:** Shipped multiple complete applications to production within 12-week timeframe, demonstrating ability to move from concept to deployed product rapidly

Technical focus: TypeScript, React, Next.js, AI SDK integration, modern full-stack development patterns in a fast-paced collaborative environment.

HOLDFAST-CIFARM | Remote

Software Engineer Intern | July 2025

Built grant application system for NOAA partnership serving aquaculture research community:

- **Proposal builder architecture:** Designed and implemented auto-saving proposal system preventing data loss during multi-session grant applications, enabling researchers to iterate on complex NOAA submissions over weeks
- **Design system foundation:** Established component library and design patterns using Tailwind CSS, creating consistent UI framework adopted across application for improved user experience
- **Full-stack development:** Built responsive proposal management interface with Next.js and TypeScript, implementing real-time draft persistence and form state management for complex multi-step workflows
- **Testing infrastructure:** Integrated Vitest testing framework ensuring reliable proposal data handling and user interaction flows

Technical stack: Next.js, TypeScript, Tailwind CSS, Vitest, pnpm in collaborative 5-person development team focused on academic research tooling.

- **Scenic Fabrication Design Shop** Brooklyn, NY
 - **3d Printing Dept Lead** 01/2024 - 05/2025
 - Developed and maintained a database of 3d prints with information relating to printers, materials used, time elapsed, quality, and issues to log successes and bottlenecks in order to maintain high quality and troubleshoot any issues that arise in production.
 - Developed production pipeline and established department in order to address need for high detailed pieces for client demands and aiding the switch from outsourcing 3d prints to sourcing in house.
 - Designed and modified existing 3d models to create high quality prints with regards to art direction.
 - Interfaced with Design, Project Management, and Scenic Department in order to create high quality prints that can be implemented into production.
 - **Scenic Artist** 05/2021 - 01/2024
 - Interfaced with Project Managers and Designers to execute designs for sculptural and scenic elements of projects for such brands as Macy's, HBO, Tiffany's, Hermès, Toys R Us, among many others.
 - Coordinated with a team of artists to ensure speedy and accurate delivery of design elements.
 - Ordered and maintained inventory of tools and materials used in the creation of scenic and sculptural elements.
 - Researched and developed new tools and materials for use in new techniques to create a more efficient and accurate product in accordance with Art Direction in service of Client's requirements.

Education & Certifications

- **Bachelor of Science in Media Arts and Animation**
Art Institute of Fort Lauderdale